

# Daniel C. Patiño

## Senior Data Scientist

danielcarvajalpatino@gmail.com | +57 311 531 5773 | Bogotá, Colombia | [LinkedIn](#) | [GitHub](#)

### PROFESSIONAL SUMMARY

Data Scientist with 6+ years in advanced modeling, deep learning, and peer-reviewed research (GANs, trading, computer vision). TensorFlow, PyTorch, and rigorous experimentation; applied experience in retail and financial markets.

### PROFESSIONAL EXPERIENCE

**Machine Learning Engineer** Mar 2023 - Present  
[Farmatodo](#) Bogotá, Colombia

- Led experimentation and evaluation for personalization and multi-channel campaign models; metric design and business-facing readouts
- Framed business questions as ML and analytics problems: segmentation, affinity, and attribution in a retail context
- Designed exploratory analyses and reproducible reporting (Jupyter/SQL) to support marketing and product decisions
- Documented findings for non-technical stakeholders; data storytelling from hypothesis to impact
- Engineered features and training datasets from behavioral and campaign signals

**Data Engineer** Apr 2022 - Mar 2023  
[Ultracom](#) (Cliente: [Bancolombia](#)) Medellín, Colombia

- Maintained and evolved components of Bancolombia's enterprise Data Warehouse (SQL/Python, batch jobs, dimensional adjustments and ingestion for business needs)
- Migrated Python pipelines to Cloudera CDP Public Cloud, aligning dependencies and deployment for more elastic scaling under load
- Deployed Django APIs on AWS (DynamoDB, Glue, S3, CloudFormation) exposing lake data and operations to internal consumers
- Applied code reviews and production deployment quality gates (testing standards) for Python services on AWS

**Data Scientist & ML Engineer** Jul 2018 - Mar 2022  
[Frontierx](#) Medellín, Colombia

- Researched and developed generative models (GANs) for synthetic trading data, publishing in indexed scientific journal
- Deployed market prediction models using TensorFlow and PyTorch in production
- Designed and deployed algorithmic trading strategies based on behavioral patterns
- Built Computer Vision models with CNNs for satellite image classification
- Researched impact of synthetic data on trading predictions (published master's thesis)

### EDUCATION

**Master's Degree in Engineering** May 2019 - Nov 2022  
[Universidad de Antioquia](#) Medellín, Colombia

- Research focus: Deep Learning under Professor Raúl Ramos Pollán's supervision
- Thesis: Impact of synthetic data (GANs) on commodities trading predictions
- Indexed scientific publication in Research in International Business and Finance (2022)

**Systems Engineering** Sep 2013 - Aug 2018  
[Universidad Industrial de Santander](#) Bucaramanga, Colombia

- Thesis: Urban socioeconomic stratification estimation using CNNs applied to satellite imagery

## PUBLICATIONS

---

Synthetic data generation with deep generative models to enhance predictive tasks in trading strategies  
Research in International Business and Finance, 2022

## TECHNICAL SKILLS

---

**Languages:**  
Python, SQL, JavaScript/TypeScript

**Databricks & Lakehouse:**  
Databricks (Lakehouse), Apache Spark / PySpark, Delta Lake, Workflows & Jobs, Unity Catalog, MLflow, Terraform (Data-bricks)

**Machine Learning & AI:**  
TensorFlow, PyTorch, scikit-learn, GANs, CNNs, Time Series, MLOps / Model governance

**Data Analysis:**  
Pandas, NumPy, Matplotlib, Jupyter, Feature Engineering

**SQL & Databases:**  
PostgreSQL, MySQL, Oracle, SQL Server

## LANGUAGES

---

|         |             |
|---------|-------------|
| Spanish | Native      |
| English | C1 Advanced |

## CERTIFICATIONS

---

TensorFlow Developer Professional Certificate  
Coursera (DeepLearning.AI) • Sep 2019

Deep Learning Specialization  
Coursera (DeepLearning.AI) • Jun 2019

Build Basic Generative Adversarial Networks (GANs)  
Coursera • May 2021

EF SET English Certificate (C1 Advanced)  
EF Education First • Feb 2023